Arunabha Majumder

Mechanical / Production Engineer | Troubleshooting, Validation, Equipment and Test Rigs

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SUMMARY

Mechanical engineer who likes being close to the hardware and fixing things until they work every time. During my PhD at Aalborg University I built and maintained test rigs, integrated motors and sensors, found and fixed faults, and documented setups so others could repeat them. I hold an MTech in Mechatronics and a BTech in Mechanical Engineering, with workshop experience in machining and assembly from my Bachelor's. I want to bring this hands-on, structured way of working into production.

KEY RESULTS

- **>25%** less energy from a shoulder exoskeleton actuator I designed and built (Best Paper Award, 2026)
- **<10% error** force estimate on a 5-bar robot, validated against a reference force sensor
- **12** participants in experiments where I collected the data with EMG and motion capture

EXPERIENCE

PhD Researcher, Mechatronics and Robotics (project VIEXO) | 2023 - Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

- Built and maintained test rigs for a shoulder exoskeleton and kept them running through repeated test sessions, finding and fixing faults in mounting, wiring, sensors and data logging.
- Integrated the motor, encoder and force sensors into the mechanical frame myself, and iterated parts in SolidWorks and 3D printing until they held up in testing. The design cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Validated a force estimate on a 5-bar parallel robot (variable-stiffness actuators, robot Jacobian) against a reference force sensor, within 10% RMS error.
- Worked in a team of 3 on an Assist-as-Needed controller: I helped with the equations, such as the Fourier fit, and defined how the GUI should behave. User muscle effort dropped by more than 15%.
- Collected data from 12 participants across 3 modes and 2 load conditions, kept clear records of each setup, and taught and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Pneumatic Artificial Muscles (Master's thesis) | 2020 - 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated and assembled pneumatic artificial muscles to an existing lab design and ran experiments on the lab's setup with valves, pressure sensors and DAQ.
- Collected the experimental data, fed it into the lab's existing neural network and used its output to design the outer PID loop of a gain-scheduled controller, reaching 0.3-0.78% position error (IEEE CONECCT 2022).

Bachelor's Capstone, Upper-Limb Rehabilitation Exoskeleton | 2018 - 2019
Siddaganga Institute of Technology, Tumakuru, India

- Designed, machined, assembled and tested it in the college workshop. Best Major Project Award and an INR 100,000 TATA Technologies grant. Also built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018).

SKILLS

- **Production-relevant:** Troubleshooting hardware, test rigs, validation against reference sensors, documentation
- **CAD and workshop:** SolidWorks (CSWA), 3D printing; machining and assembly (Bachelor's)
- **Equipment:** Motors, encoders, force/torque and pressure sensors, pneumatic valves, DAQ, Arduino
- **Software:** MATLAB/Simulink, Python, LabVIEW (basic)

- **Drawing:** Diploma in Fine Arts; hand sketches, technical illustrations and clear figures for work instructions and reports
- **Languages:** English (C1, fluent), Bengali (native), Hindi (conversational)

EDUCATION

PhD, Mechatronics and Robotics (defence expected in 2027) | 2023 - Present
Aalborg University, Denmark

MTech, Mechatronics | 2020 - 2022
AcSIR - CSIR-CMERI, Durgapur, India

BTech, Mechanical Engineering | 2015 - 2019
Siddaganga Institute of Technology (VTU), India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer.
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.

REFERENCES

Available upon request.